

The F-Statistic

i) Pairwise Difference/Contrast

$$ATE_{1,2} = \bar{Y}_1 - \bar{Y}_2$$

$$\widehat{ATE}_{1,2} = \widehat{\bar{Y}}_1 - \widehat{\bar{Y}}_2$$

ii) Effect relative to benchmark

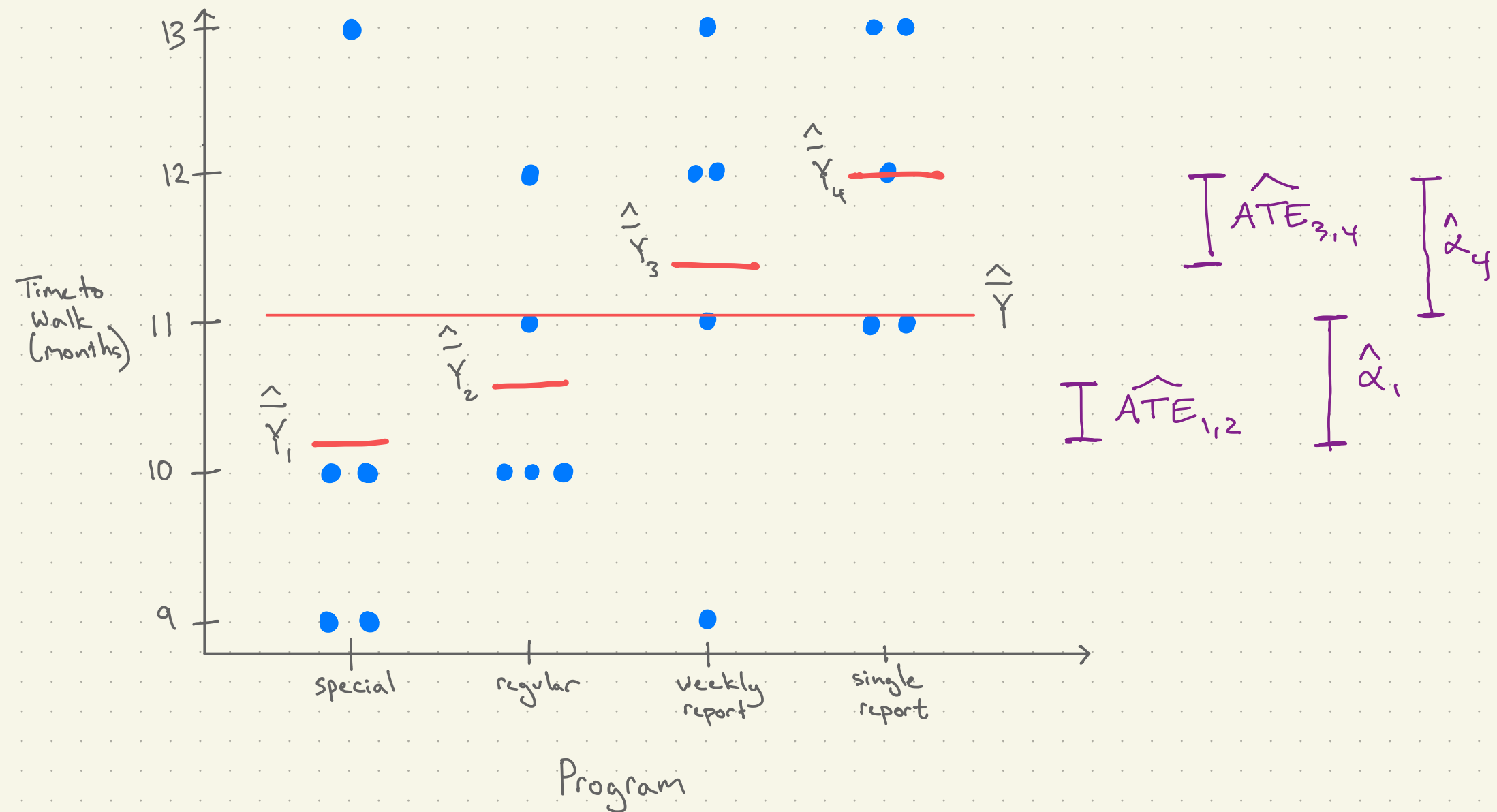
for 1st group:

$$\alpha_1 = \frac{1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{4n} \sum_{i=1}^n \sum_{j=1}^4 Y_i(j)$$

$$\widehat{\alpha}_1 = \frac{1}{n_1} \sum_{i \in g_1} Y_i - \frac{1}{n} \sum_{i=1}^n Y_i = \widehat{\bar{Y}}_1 - \widehat{\bar{Y}}$$

↑
indices of
group 1

Visualizing Statistics



Steps of a Hypothesis Test

- 1) Choose an alpha-level (α) / decision boundary.
 - ↳ $0 \leq \alpha \leq 1$
 - ↳ $\alpha \geq P_{H_0}(\text{Reject } H_0) \leftarrow \text{Type I error}$
- 2) Define a random data generating process. H_A .
- 3) Determine meaningful null hypothesis, H_0 , and alt.,
- 4) Determine a test statistic, T (which is a RV).
- 5) Derive / approximate distribution of T .
- 6) Decide whether or not to reject null, H_0 .

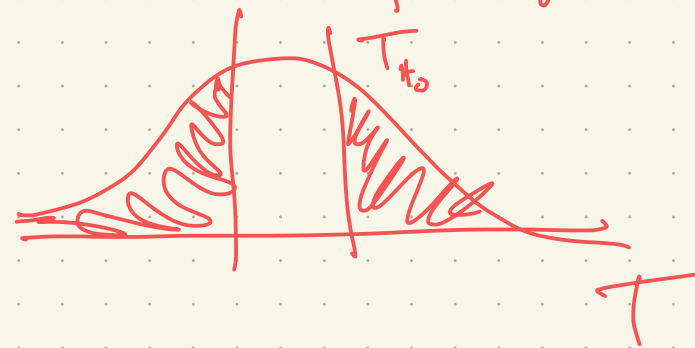
$Y_i(1)$	$Y_i(2)$	$Y_i(3)$	$Y_i(4)$
9	?	?	?
?	11	?	?
?	?	11	?
?	?	?	13

→

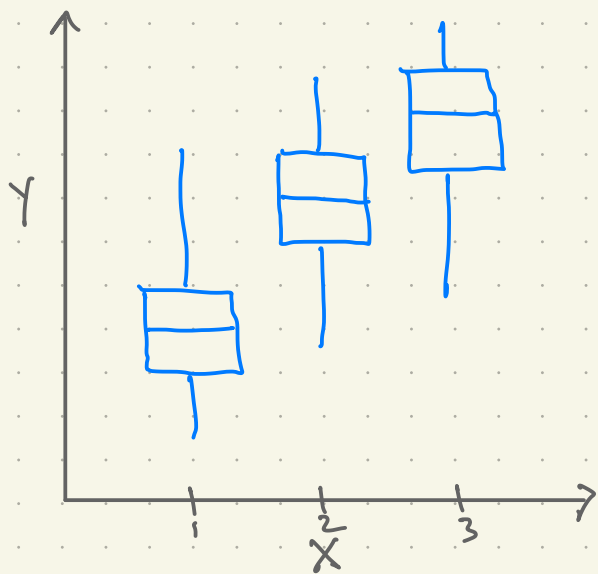
$Y_i(1)$	$Y_i(2)$	$Y_i(3)$	$Y_i(4)$
9	9	9	9
11	11	11	11
11	11	11	11
13	13	13	13

RBI

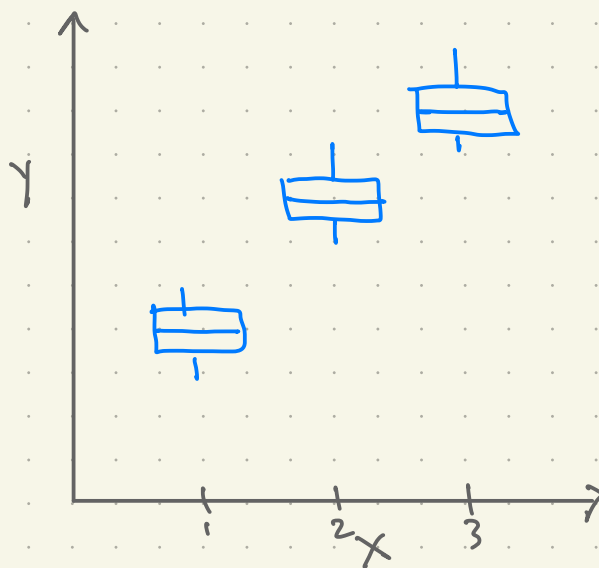
- $\alpha = .05$
- Potential Outcomes
- $H_0: Y_i(1) = Y_i(2) = Y_i(3) = Y_i(4)$
- $\hat{\alpha}_1, t = \frac{\hat{\beta}_1}{SE}, T_{JACOB}$
- Impute Null Schedule + Draws
(Shuffle group membership)
- if $pval \leq \alpha$
↳ reject H_0



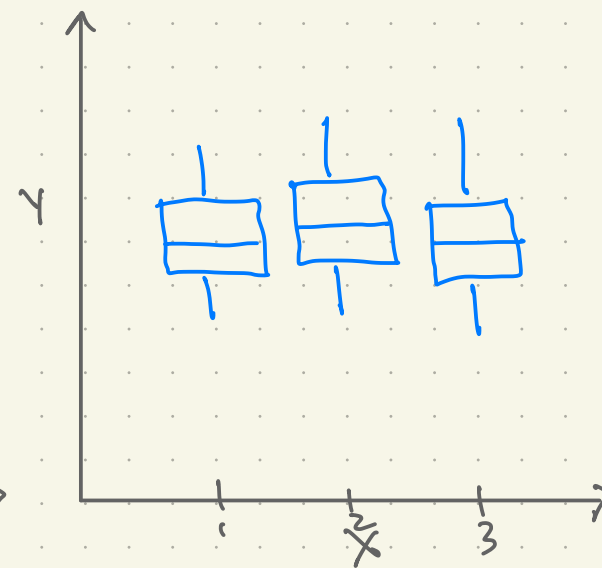
Ⓐ



Ⓑ



Ⓒ



Q Rank the plots from strongest to weakest suggestion of a causal effect of X on Y.

Q Create a statistic to measure this effect.

Desiderata:

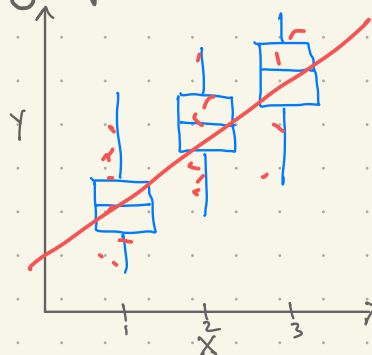
- ▣ function of data, Y_i , or statistics ($\hat{Y}_i, n, m, \max, \text{etc}$)
- ▣ outputs a single number
- ▣ is large when the effect is large.

Some Statistics

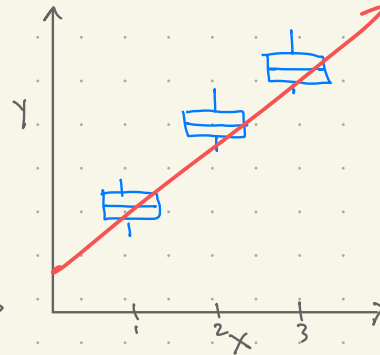
① T
JACOB

$$\frac{\sum_{i=1}^n (y_i - \hat{y}_{j(i)})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

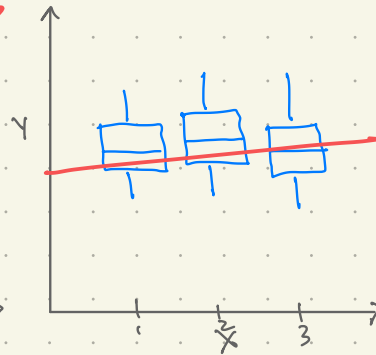
$j(i)$: group of i th obs.



.82



.17



1

②

$$t = \frac{\hat{B}_1}{SE_{\hat{B}_1}}$$

1.3

2.7

≈ 0

③

$$F = \frac{MS_{12}}{MS_w}$$

1.2

2.3

1

The F-statistic

"The variability of the group averages
relative to the
average variability within each group"

$$\frac{A}{B}$$

B. "average variability within each group." $j \in \{1, 2, \dots, J\}$ group #

$$s_j^2 = \frac{1}{n_j - 1} \sum_{i \in g_j} (Y_i - \hat{Y}_{j(i)})^2$$

$$\frac{1}{J} \sum_{j=1}^J s_j^2$$

Let's instead do a weighted avg: $\frac{1}{w_1 + w_2 + w_3 + w_4} \sum_{i=1}^4 w_i X_i$

$$\frac{1}{n - J} \sum_{j=1}^J (n_j - 1) s_j^2$$

$$\leftarrow \left[\sum_{j=1}^J (n_j - 1) \right] = \sum_{j=1}^J n_j - \sum_{j=1}^J 1 = n - J$$

$$= \frac{1}{n - J} \sum_{i=1}^n (Y_i - \hat{Y}_{j(i)})^2$$

= MS_w "mean squared within"

A. "The variability of the group averages"

$$\frac{1}{J-1} \sum_{j=1}^J (\bar{\hat{Y}}_j - \bar{\hat{Y}})^2$$

Lets weight each contribution to the variance by the group size: n_j

$$\frac{1}{J-1} \sum_{j=1}^J n_j (\bar{\hat{Y}}_j - \bar{\hat{Y}})^2$$

= MS_B "mean squared between"

The F-statistic

"The variability of the group averages
relative to the
average variability within each group."

$$\frac{A}{B} = \frac{MS_B}{MS_W} = \frac{\frac{1}{J-1} \sum_{j=1}^J n_j (\bar{\hat{Y}}_j - \bar{Y})^2}{\frac{1}{n-J} \sum_{i=1}^n (Y_i - \bar{\hat{Y}}_{j(i)})^2}$$

